📋 CMDB Inventory — Documentation Complète des URLs

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1. Vue d'Ensemble des Modules

Module	Base URL	Description	Auth Requise
Dashboard	`/`	Vitrine publique, statistiques	❌ Non
Admin Métier	`/admin/`	Interface techniciens/gestionnaires	✅ Oui
Scanner	`/admin/scanner/`	Scan QR/Code-barres	✅ Oui
Scan Public	`/scan/<uuid>/`	Fiche asset publique (terrain)	❌ Non
API Inventory	`/api/v1/inventory/`	Gestion assets	✅ Oui
API Maintenance	`/api/v1/maintenance/`	Tickets, réparations	✅ Oui
API Scanner	`/api/v1/scanner/`	Résolution QR, logs	⚠️ Mixte
API Printer	/api/printer/	Impression, templates, printers<	✅ Oui
API Stock	`/api/v1/stock/`	Articles atelier, mouvements	✅ Oui

Module	Base URL	Description	Auth Requise
API Auth	`/api/v1/auth/`	Token, utilisateurs	✅ Oui
Django Admin	`/django-admin/`	Administration système	✅ Superuser

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2. URLs Publiques (Sans Authentication)

2.1 Dashboard & Pages Publiques

URL	Méthode	Vue/Template	Description	Usage
`/`	GET	`dashboard.html`	Dashboard vitrine publique	Stats, KPIs, navigation
`/scan/<uuid>/`	GET	`public/scan_result.html`	Fiche asset publique (QR scan)	Techniciens terrain sans compte
`/admin/login/`	GET/POST	`admin_login.html`	Authentification Token	Login techniciens
`/media/qr_codes/<file>`	GET	-	Images QR Code (static)	Téléchargement étiquettes

2.2 Détails — `/scan/<uuid>/`

Élément	Détail
Paramètre	`uuid` = UUID du QR Code ou Serial Number
Template	`public/scan_result.html`
View	`scanner.views.public_scan_result`
Données affichées	Nom, S/N, photo, statut, localisation, assigné, garantie, tickets ouverts
Actions	Télécharger QR, voir tickets
Auth	❌ Aucune (accessible sans compte)
Use Case	Technicien terrain scanne QR avec smartphone → fiche asset sans login

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3. URLs Interface Métier (/admin/)

3.1 Layout & Navigation

URL	Méthode	Template	Description	Composants
`/admin/`	GET	`admin_base.html` + dashboard	Dashboard métier (après login)	Sidebar, Navbar, KPIs
`/admin/assets/`	GET	`admin/assets/list.html`	Liste assets + filtres	Tableau, filtres, pagination
`/admin/assets/new/`	GET/POST	`admin/assets/form.html`	Création asset	Formulaire complet

URL	Méthode	Template	Description	Composants
`/admin/assets/<id>/`	GET	`admin/assets/detail.html`	Fiche détail asset	4 onglets (Infos, Mouvements, Tickets, QR)
`/admin/assets/<id>/edit/`	GET/PUT	`admin/assets/form.html`	Édition asset	Formulaire pré-rempli
`/admin/tickets/`	GET	`admin/tickets/list.html`	Tickets Kanban + tableau	Drag-drop, statuts
`/admin/tickets/new/`	GET/POST	`admin/tickets/form.html`	Création ticket	Formulaire + sélection asset
`/admin/tickets/<id>/`	GET	`admin/tickets/detail.html`	Détail ticket	Workflow, commentaires, pièces
`/admin/scanner/`	GET	`admin/scanner/index.html`	Interface scan (USB/Webcam)	ZXing, buffer USB, historique
`/admin/stock/`	GET	`admin/stock/list.html`	Stock atelier + alertes	Tableau, badges stock
`/admin/stock/<id>/`	GET	`admin/stock/detail.html`	Fiche article + mouvements	Timeline, stats
`/admin/stock/movement/`	GET/POST	`admin/stock/movement.html`	Entrée/Sortie rapide	Modal, lien ticket
`/admin/search/`	GET	`admin/search/results.html`	Recherche globale	Assets + Tickets + Stock
`/admin/scan-print/`	GET	`admin/scan_print/landing.html`	Hub Scan & Print	Stats, activité récente, actions rapides
`/admin/scanner/print/`	GET/POST	`admin/scanner/print.html`	Impression étiquettes	Templates, imprimantes, PDF batch
`/admin/scanner/templates/`	GET/POST	`admin/scanner/templates.html`	Gestion templates	CRUD PrintTemplate
`/admin/scanner/printers/`	GET/POST	`admin/scanner/printers.html`	Gestion imprimantes	CRUD Printer (USB/BT/WiFi)
`/admin/scanner/jobs/`	GET	`admin/scanner/jobs.html`	Historique jobs	PrintJob status, download PDF
`/admin/scanner/logs/`	GET	`admin/scanner/logs.html`	Logs audit impressions	PrintLog, filtres, export

3.2 Détails — Module Assets

URL	Méthode	API Consommée	Description	Permissions
`/admin/assets/`	GET	`GET /api/v1/inventory/assets/`	Liste paginée + filtres	Technicien+
`/admin/assets/new/`	GET/POST	`POST /api/v1/inventory/assets/`	Création + QR auto	Gestionnaire+
`/admin/assets/<id>/`	GET	`GET /api/v1/inventory/assets/<id>/`	Fiche détail 4 onglets	Technicien+
`/admin/assets/<id>/edit/`	GET/PUT	`PUT /api/v1/inventory/assets/<id>/`	Édition asset	Gestionnaire+
`/admin/assets/<id>/move/`	POST	`POST /api/v1/inventory/assets/<id>/move/`	Changer localisation	Technicien+
`/admin/assets/<id>/assign/`	POST	`POST /api/v1/inventory/assets/<id>/assign/`	Assigner utilisateur	Gestionnaire+
`/admin/assets/<id>/retire/`	POST	`POST /api/v1/inventory/assets/<id>/retire/`	Archiver asset	Gestionnaire+

3.3 Détails — Module Scanner

URL	Méthode	API Consommée	Description	Permissions
`/admin/scanner/`	GET	`GET /api/v1/scanner/scan/<uuid>/`	Interface scan USB/Webcam	Technicien+
`/admin/scanner/print/`	GET/POST	`POST /api/v1/scanner/print-labels/`	Impression étiquettes batch	Technicien+
`/admin/scanner/templates/`	GET/POST	API PrintTemplate	Gestion templates	Admin+
`/admin/scanner/printers/`	GET/POST	API Printer	Configuration imprimantes	Admin+

URL	Méthode	API Consommée	Description	Permissions
`/admin/scanner/jobs/`	GET	API PrintJob	Historique jobs	Technicien+
`/admin/scanner/logs/`	GET	API PrintLog	Audit impressions	Admin+

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4. API REST Inventory

4.1 Assets

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/inventory/assets/`	GET	Liste assets paginée	`page`, `page_size`, `search`, `category`, `brand`, `status`, `location`	`{count, next, previous, results[]}`	✅ Token
`/api/v1/inventory/assets/`	POST	Créer asset	`name`, `serial_number`, `category`, `brand`, `status`	`{id, uuid, name, ...}`	✅ Token
`/api/v1/inventory/assets/<id>/`	GET	Détail asset	-	`{id, name, serial_number, category, brand, location, assigned_to, status, ...}`	✅ Token
`/api/v1/inventory/assets/<id>/`	PUT	Mettre à jour asset	Tous champs	`{id, name, ...}`	✅ Token
`/api/v1/inventory/assets/<id>/`	DELETE	Supprimer asset	-	`204 No Content`	✅ Admin
`/api/v1/inventory/assets/<id>/move/`	POST	Déplacer asset	`location_id`	`{status, new_location}`	✅ Token
`/api/v1/inventory/assets/<id>/assign/`	POST	Assigner asset	`user_id`	`{status, assigned_to}`	✅ Token
`/api/v1/inventory/assets/<id>/retire/`	POST	Archiver asset	-	`{status, retired_at}`	✅ Token
`/api/v1/inventory/assets/warranty-expiring/`	GET	Garanties expirant bientôt	`days` (default 30)	`[{asset, warranty_end}]`	✅ Token
`/api/v1/inventory/assets/by-status/`	GET	Répartition par statut	-	`[{status, count}]`	✅ Token
`/api/v1/inventory/assets/by-category/`	GET	Répartition par catégorie	-	`[{category, count}]`	✅ Token
`/api/v1/inventory/assets/export/csv/`	GET	Export CSV	Filtres	`text/csv`	✅ Token

4.2 Catégories, Marques, Locations, Tags

Endpoint	Méthode	Description	Auth
`/api/v1/inventory/category/`	GET/POST	Liste/Créer catégories	✅ Token
`/api/v1/inventory/category/<id>/`	GET/PUT/DELETE	Détail/Maj/Suppr catégorie	✅ Token

Endpoint	Méthode	Description	Auth
`/api/v1/inventory/brand/`	GET/POST	Liste/Créer marques	✅ Token
`/api/v1/inventory/brand/<id>/`	GET/PUT/DELETE	Détail/Maj/Suppr marque	✅ Token
`/api/v1/inventory/location/`	GET/POST	Liste/Créer localisations	✅ Token
`/api/v1/inventory/location/<id>/`	GET/PUT/DELETE	Détail/Maj/Suppr location	✅ Token
`/api/v1/inventory/tag/`	GET/POST	Liste/Créer tags	✅ Token
`/api/v1/inventory/tag/<id>/`	GET/PUT/DELETE	Détail/Maj/Suppr tag	✅ Token

4.3 Mouvements Assets

Endpoint	Méthode	Description	Params	Auth
`/api/v1/inventory/asset-movement/`	GET	Historique mouvements	`asset`, `location`, `date_from`, `date_to`	✅ Token
`/api/v1/inventory/asset-movement/`	POST	Créer mouvement	`asset_id`, `from_location`, `to_location`, `movement_type`	✅ Token
`/api/v1/inventory/asset-movement/<id>/`	GET	Détail mouvement	-	✅ Token
`/api/v1/inventory/asset-movement/<id>/`	DELETE	Supprimer mouvement	-	✅ Admin

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5. API REST Maintenance

5.1 Tickets

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/maintenance/tickets/`	GET	Liste tickets	`page`, `status`, `priority`, `assignee`, `asset`	`{count, results[]}`	✅ Token
`/api/v1/maintenance/tickets/`	POST	Créer ticket	`subject`, `description`, `asset_id`, `priority`, `category`	`{id, subject, status}`	✅ Token
`/api/v1/maintenance/tickets/<id>/`	GET	Détail ticket	-	`{id, subject, description, status, priority, asset, assignee, ...}`	✅ Token
`/api/v1/maintenance/tickets/<id>/`	PATCH	Mettre à jour ticket	Champs partiels	`{id, ...}`	✅ Token
`/api/v1/maintenance/tickets/<id>/`	DELETE	Supprimer ticket	-	`204 No Content`	✅ Admin

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/maintenance/tickets/<id>/transition/`	PATCH	Changer statut	`to_status`	`{status, history}`	✅ Token
`/api/v1/maintenance/tickets/<id>/assign/`	POST	Assigner ticket	`assignee_id`	`{assignee, status}`	✅ Token
`/api/v1/maintenance/tickets/<id>/comments/`	GET	Liste commentaires	-	`[{id, content, author, created_at, is_internal}]`	✅ Token
`/api/v1/maintenance/tickets/<id>/comments/`	POST	Ajouter commentaire	`content`, `is_internal`	`{id, content, author}`	✅ Token
`/api/v1/maintenance/tickets/<id>/parts/`	GET	Pièces consommées	-	`[{item, quantity, unit_price, total_price}]`	✅ Token
`/api/v1/maintenance/tickets/<id>/parts/`	POST	Ajouter pièce	`item_id`, `quantity`	`{item, quantity, total}`	✅ Token
`/api/v1/maintenance/tickets/overdue/`	GET	Tickets en retard	-	`[{id, subject, due_date}]`	✅ Token
`/api/v1/maintenance/tickets/stats/`	GET	Statistiques tickets	-	`{open, in_progress, resolved_30d, overdue}`	✅ Token
`/api/v1/maintenance/tickets/export/csv/`	GET	Export CSV	Filtres	`text/csv`	✅ Token

5.2 Workflows & Statuts

Statut	Code	Couleur	Transitions Autorisées
Ouvert	`open`	🔵 Bleu	→ assigned, in_progress, closed
Assigné	`assigned`	🟣 Violet	→ in_progress, open, closed
En cours	`in_progress`	🟠 Orange	→ waiting_parts, resolved, assigned
Attente pièces	`waiting_parts`	🟡 Ambre	→ in_progress, resolved
Résolu	`resolved`	🟢 Vert	→ closed, in_progress
Fermé	`closed`	⚫ Gris	→ open (réouverture)
Annulé	`cancelled`	🔴 Rouge	-

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6. API REST Scanner

6.1 Scan & Résolution

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/scanner/scan/<uuid>/`	GET	Résoudre QR/Code-barres	`uuid` = UUID, S/N, ou code interne	`{id, name, serial_number, status, location, ...}`	❌ Public

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/scanner/stats/`	GET	Statistiques scans	`date_from`, `date_to`	`{scans_today, scans_week, top_assets}`	✅ Token
`/api/v1/scanner/recent-activity/`	GET	Activité récente	`limit` (default 10)	`[{type, asset, user, timestamp}]`	✅ Token
`/api/v1/scanner/logs/`	GET	Historique scans	`asset`, `user`, `date_from`, `date_to`	`[{id, asset, scanned_by, timestamp, ip}]`	✅ Token

6.2 QR Code Management

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/scanner/assets/<id>/regen-qr/`	POST	Régénérer QR Code	`asset_id` (URL)	`{uuid, url, image}`	✅ Token
`/api/v1/scanner/assets/<id>/qr/`	GET	Récupérer QR info	`asset_id` (URL)	`{uuid, code, image_url, scanned_count}`	✅ Token
`/api/v1/scanner/batch-generate/`	POST	Générer QR batch	`asset_ids[]`	`{success, failed, count}`	✅ Token

6.3 Impression Étiquettes

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/scanner/print-label/`	GET	PDF single asset	`asset_id`, `format`, `copies`	`application/pdf`	✅ Token
`/api/v1/scanner/print-labels/`	POST	PDF batch assets	`asset_ids[]`, `template_id`, `printer_id`, `copies`	`application/pdf`	✅ Token
`/api/v1/scanner/templates/`	GET/POST	CRUD templates	-	`[{id, name, format, ...}]`	✅ Token
`/api/v1/scanner/templates/<id>/`	GET/PUT/DELETE	Détail template	-	`{id, name, format, ...}`	✅ Token
`/api/v1/scanner/printers/`	GET/POST	CRUD imprimantes	-	`[{id, name, connection_type, ...}]`	✅ Token
`/api/v1/scanner/printers/<id>/`	GET/PUT/DELETE	Détail imprimante	-	`{id, name, connection_type, ...}`	✅ Token
`/api/v1/scanner/printers/<id>/test/`	POST	Tester imprimante	-	`application/pdf` (test)	✅ Token
`/api/v1/scanner/print-jobs/`	GET/POST	CRUD jobs	`status`, `created_by`	`[{id, status, asset_count, ...}]`	✅ Token
`/api/v1/scanner/print-jobs/<id>/`	GET	Détail job	-	`{id, status, pdf_url, ...}`	✅ Token
`/api/v1/scanner/print-jobs/<id>/download/`	GET	Télécharger PDF	-	`application/pdf`	✅ Token
`/api/v1/scanner/print-jobs/<id>/cancel/`	POST	Annuler job	-	`{status: cancelled}`	✅ Token
`/api/v1/scanner/print-logs/`	GET	Logs audit	`asset`, `user`, `date_from`, `date_to`	`[{id, asset, printed_by, timestamp}]`	✅ Token

7. API REST Printer ⭐ NOUVEAU

7.1 Vue d'Ensemble Module Printer

Composant	Description	Usage
PrintTemplate	Modèles d'étiquettes (formats, layouts)	30x20mm, 50x30mm, 70x40mm, etc.
Printer	Configuration imprimantes (USB/BT/WiFi)	MUNBYN, Samsung, Zebra
PrintJob	Jobs d'impression batch (PDF)	Multi-assets, tracking
PrintLog	Audit des impressions (traçabilité)	Qui, quand, quoi
PrintLabel	Génération étiquettes unitaires	Single asset

7.2 Endpoints API Printer

Endpoint	Méthode	View/ViewSet	Description	Params	Response	Auth
`/api/printer/`	GET	`APIRootView`	Racine API Printer	-	`{endpoints}`	✅ Token
`/api/printer/printers/`	GET	`PrinterViewSet`	Liste imprimantes	`is_active`, `connection_type`	`[{id, name, connection_type, ...}]`	✅ Token
`/api/printer/printers/`	POST	`PrinterViewSet`	Créer imprimante	`name`, `connection_type`, `ip_address`, `device_path`	`{id, name, ...}`	✅ Admin
`/api/printer/printers/<id>/`	GET	`PrinterViewSet`	Détail imprimante	-	`{id, name, connection_type, config, ...}`	✅ Token
`/api/printer/printers/<id>/`	PUT	`PrinterViewSet`	Mettre à jour imprimante	Champs partiels	`{id, name, ...}`	✅ Admin
`/api/printer/printers/<id>/`	DELETE	`PrinterViewSet`	Supprimer imprimante	-	`204 No Content`	✅ Admin
`/api/printer/templates/`	GET	`PrintTemplateViewSet`	Liste templates	`format`, `is_default`	`[{id, name, format, width_mm, height_mm}]`	✅ Token
`/api/printer/templates/`	POST	`PrintTemplateViewSet`	Créer template	`name`, `format`, `width_mm`, `height_mm`, `layout`	`{id, name, ...}`	✅ Admin
`/api/printer/templates/<id>/`	GET	`PrintTemplateViewSet`	Détail template	-	`{id, name, format, layout_config}`	✅ Token
`/api/printer/templates/<id>/`	PUT	`PrintTemplateViewSet`	Mettre à jour template	Champs partiels	`{id, name, ...}`	✅ Admin
`/api/printer/templates/<id>/`	DELETE	`PrintTemplateViewSet`	Supprimer template	-	`204 No Content`	✅ Admin
`/api/printer/jobs/`	GET	`PrintJobViewSet`	Liste jobs impression	`status`, `created_by`, `date_from`, `date_to`	`[{id, status, asset_count, pdf_url}]`	✅ Token
`/api/printer/jobs/`	POST	`PrintJobViewSet`	Créer job impression	`asset_ids[]`, `template_id`, `printer_id`, `copies`	`{id, status, uuid}`	✅ Token
`/api/printer/jobs/<id>/`	GET	`PrintJobViewSet`	Détail job	-	`{id, status, pdf_url, completed_at}`	✅ Token
`/api/printer/jobs/<id>/download/`	GET	`PrintJobViewSet`	Télécharger PDF	-	`application/pdf`	✅ Token
`/api/printer/jobs/<id>/cancel/`	POST	`PrintJobViewSet`	Annuler job	-	`{status: cancelled}`	✅ Token
`/api/printer/labels/`	GET	`PrintLabelViewSet`	Liste étiquettes	`asset_id`, `status`	`[{id, asset, qr_code, status}]`	✅ Token
`/api/printer/labels/`	POST	`PrintLabelViewSet`	Générer étiquette	`asset_id`, `template_id`, `copies`	`{id, pdf_url, status}`	✅ Token

Endpoint	Méthode	View/ViewSet	Description	Params	Response	Auth
`/api/printer/labels/status/`	GET	`PrintLabelViewSet`	Statut imprimantes	-	`{printers: [{name, status, online}]}`	✅ Token
`/api/printer/print-labels/`	POST	`print_labels`	Impression directe	`asset_ids[]`, `template_id`, `copies`	`application/pdf`	✅ Token
`/django-admin/inventory/asset/<id>/generate_qrcode/`	POST	`generate_qrcode_view`	Générer QR Code asset	`asset_id` (URL)	`{uuid, qr_url, image}`	✅ Admin
`/django-admin/inventory/asset/<id>/print_label/`	GET	`print_label_pdf_view`	PDF étiquette single	`asset_id` (URL), `format`, `copies`	`application/pdf`	✅ Admin

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7.3 Modèles Printer

PrintTemplate

Champ	Type	Description	Exemple
`name`	CharField	Nom du template	"Standard Laptop 50x30"
`format`	CharField	Format étiquette	"50x30", "70x40", "80x50"
`width_mm`	Integer	Largeur (mm)	50
`height_mm`	Integer	Hauteur (mm)	30
`qr_size_mm`	Integer	Taille QR Code (mm)	20
`qr_position_x_mm`	Integer	Position X QR (mm)	5
`qr_position_y_mm`	Integer	Position Y QR (mm)	5
`font_size_title`	Integer	Taille police titre	10
`font_size_text`	Integer	Taille police texte	8
`show_serial`	Boolean	Afficher S/N	True
`show_category`	Boolean	Afficher catégorie	True
`show_location`	Boolean	Afficher localisation	True
`is_default`	Boolean	Template par défaut	False
`is_active`	Boolean	Template actif	True

Printer

Champ	Type	Description	Exemple
`name`	CharField	Nom imprimante	"MUNBYN RW403B - Atelier"
`connection_type`	CharField	Type connexion	"usb", "bluetooth", "wifi", "network"
`ip_address`	GenericIP	Adresse IP (WiFi/Ethernet)	"192.168.1.100"

Champ	Type	Description	Exemple
`port`	Integer	Port réseau	9100
`mac_address`	CharField	Adresse MAC (Bluetooth)	"00:11:22:33:44:55"
`device_path`	CharField	Chemin device (USB)	"/dev/usb/lp0"
`dpi`	Integer	Résolution	203
`speed`	Integer	Vitesse (1-5)	3
`density`	Integer	Densité (1-15)	10
`paper_width_mm`	Integer	Largeur papier (mm)	50
`paper_height_mm`	Integer	Hauteur papier (mm)	30
`is_active`	Boolean	Imprimante active	True
`is_default`	Boolean	Imprimante par défaut	True
`last_used_at`	DateTime	Dernière utilisation	2026-03-26 14:30

PrintJob

Champ	Type	Description	Exemple
`uuid`	UUID	Identifiant unique job	"abc123-def456-789"
`created_by`	ForeignKey	Utilisateur créateur	User #5
`asset_ids`	JSONField	Liste IDs assets	"[210, 211, 212]"
`template`	ForeignKey	Template utilisé	PrintTemplate #1
`printer`	ForeignKey	Imprimante cible	Printer #1
`copies`	Integer	Nombre de copies	1
`status`	CharField	Statut job	"pending", "processing", "completed", "failed"
`pdf_file`	FileField	Fichier PDF généré	"print_jobs/2026/03/job_abc.pdf"
`error_message`	TextField	Erreur (si failed)	"Printer offline"
`started_at`	DateTime	Début traitement	2026-03-26 14:30
`completed_at`	DateTime	Fin traitement	2026-03-26 14:31
`created_at`	DateTime	Création job	2026-03-26 14:30

PrintLog

Champ	Type	Description	Exemple
`job`	ForeignKey	Job associé	PrintJob #12
`asset`	ForeignKey	Asset imprimé	Asset #210
`printed_by`	ForeignKey	Utilisateur (qui)	User #5
`printer_name`	CharField	Nom imprimante	"MUNBYN RW403B"
`template_name`	CharField	Nom template	"Standard 50x30"
`printed_at`	DateTime	Timestamp impression	2026-03-26 14:31
`ip_address`	GenericIP	IP utilisateur	"192.168.1.50"
`user_agent`	TextField	User agent navigateur	"Mozilla/5.0..."

7.4 Formats d'Étiquettes Supportés

Format	Dimensions (mm)	Usage Recommandé	QR Size
`30x20`	30 × 20	Petit matériel (câbles, adaptateurs)	15mm
`50x30`	50 × 30	**Standard** (laptops, écrans)	20mm
`70x40`	70 × 40	Grand matériel (serveurs, baies)	25mm
`80x50`	80 × 50	Rack, armoires, équipements	30mm
`100x50`	100 × 50	Étiquettes industrielles	35mm

7.5 Statuts PrintJob

Statut	Code	Couleur	Description	Transitions
En attente	`pending`	⚪ Gris	Job créé, pas encore traité	→ processing, cancelled
En cours	`processing`	🔵 Bleu	Génération PDF en cours	→ completed, failed
Terminé	`completed`	🟢 Vert	PDF généré, prêt à imprimer	-
Échoué	`failed`	🔴 Rouge	Erreur génération/impression	→ pending (retry)
Annulé	`cancelled`	⚫ Noir	Job annulé par utilisateur	-

7.6 Workflow Impression

```
┌──────────────────────────────────────────────────────────────────────┐
│              WORKFLOW IMPRESSION ÉTIQUETTES                          │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  1. POST /api/printer/jobs/                                         │
│     Body: {                                                         │
│       "asset_ids": [210, 211, 212],                                 │
│       "template_id": 1,                                             │
│       "printer_id": 1,                                              │
│       "copies": 1                                                   │
│     }                                                               │
│         │                                                            │
│         ▼                                                            │
│  2. PrintJob créé (status: pending)                                 │
│     └── UUID généré                                                 │
│         │                                                            │
│         ▼                                                            │
│  3. Celery task: generate_print_job_pdf.delay(job_id)               │
│     └── Status: processing                                          │
│         │                                                            │
│         ▼                                                            │
│  4. PDF généré (ReportLab)                                          │
│     └── media/print_jobs/YYYY/MM/job_<uuid>.pdf                     │
│         │                                                            │
│         ▼                                                            │
│  5. PrintLog enregistré (par asset)                                 │
│     └── Audit: qui, quand, quelle imprimante                        │
│         │                                                            │
│         ▼                                                            │
│  6. Status: completed                                               │
│     └── PDF prêt à télécharger/imprimer                             │
│         │                                                            │
│         ▼                                                            │
│  7. GET /api/printer/jobs/<id>/download/                            │
│     └── application/pdf                                             │
│         │                                                            │
│         ▼                                                            │
│  8. Impression physique (MUNBYN/Samsung)                            │
│     └── Étiquettes collées sur assets                               │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

7.7 Exemples de Requêtes

Créer Imprimante

```
curl -X POST http://localhost:8000/api/printer/printers/ \
  -H "Authorization: Token YOUR_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{
    "name": "MUNBYN RW403B - Atelier",
    "connection_type": "usb",
    "device_path": "/dev/usb/lp0",
    "dpi": 203,
    "speed": 3,
    "density": 10,
    "paper_width_mm": 50,
    "paper_height_mm": 30,
    "is_default": true
  }'
```

Créer Template

```
curl -X POST http://localhost:8000/api/printer/templates/ \
  -H "Authorization: Token YOUR_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{
    "name": "Standard Laptop 50x30",
    "format": "50x30",
    "width_mm": 50,
    "height_mm": 30,
    "qr_size_mm": 20,
    "qr_position_x_mm": 5,
    "qr_position_y_mm": 5,
    "font_size_title": 10,
    "font_size_text": 8,
    "show_serial": true,
    "show_category": true,
    "show_location": true,
    "is_default": true
  }'
```

Créer Job Impression

```
curl -X POST http://localhost:8000/api/printer/jobs/ \
  -H "Authorization: Token YOUR_TOKEN" \
  -H "Content-Type: application/json" \
  -d '{
    "asset_ids": [210, 211, 212, 213, 214],
    "template_id": 1,
    "printer_id": 1,
    "copies": 1
  }'
```

```
# Response:
{
  "id": 45,
  "uuid": "abc123-def456-789",
  "status": "pending",
  "asset_count": 5,
  "message": "Job en cours de traitement"
}
```

Télécharger PDF

```
curl -X GET http://localhost:8000/api/printer/jobs/45/download/ \
  -H "Authorization: Token YOUR_TOKEN" \
  --output labels_20260326.pdf
```

Générer QR Code Single Asset

```
curl -X POST http://localhost:8000/api/printer/django-
admin/inventory/asset/210/generate_qrcode/ \
  -H "Authorization: Token YOUR_TOKEN"

# Response:
{
  "uuid": "abc123-def456",
  "qr_url": "/media/qr_codes/2026/03/qr_asset_210_abc.png",
  "image": "http://localhost:8000/media/qr_codes/2026/03/qr_asset_210_abc.png"
}
```

Imprimer Étiquette Single Asset

```
curl -X GET "http://localhost:8000/api/printer/django-
admin/inventory/asset/210/print_label/?format=50x30&copies=1" \
  -H "Authorization: Token YOUR_TOKEN" \
  --output label_210.pdf
```

---

8. API REST Stock

8.1 Articles Stock

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/stock/items/`	GET	Liste articles	`page`, `search`, `type`, `category`	`{count, results[]}`	✅ Token

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/stock/items/`	POST	Créer article	`reference`, `name`, `item_type`, `quantity`, `unit_price`	`{id, reference, ...}`	✅ Token
`/api/v1/stock/items/<id>/`	GET	Détail article	-	`{id, reference, name, quantity, min_quantity, ...}`	✅ Token
`/api/v1/stock/items/<id>/`	PUT	Mettre à jour article	Champs partiels	`{id, ...}`	✅ Token
`/api/v1/stock/items/<id>/`	DELETE	Supprimer article	-	`204 No Content`	✅ Admin
`/api/v1/stock/items/low-stock/`	GET	Alertes rupture	-	`[{id, name, quantity, min_quantity}]`	✅ Token
`/api/v1/stock/items/stats/`	GET	Statistiques stock	-	`{total_items, total_value, low_stock, out_of_stock}`	✅ Token
`/api/v1/stock/items/<id>/add-stock/`	POST	Entrée stock	`quantity`, `reason`, `reference_doc`	`{status, new_quantity}`	✅ Token
`/api/v1/stock/items/<id>/remove-stock/`	POST	Sortie stock	`quantity`, `reason`, `ticket_id`	`{status, new_quantity}`	✅ Token
`/api/v1/stock/items/export/csv/`	GET	Export CSV	Filtres	`text/csv`	✅ Token

8.2 Mouvements Stock

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/stock/movements/`	GET	Historique mouvements	`item`, `type`, `ticket`, `date_from`, `date_to`	`[{id, item, type, quantity, reason, ...}]`	✅ Token
`/api/v1/stock/movements/`	POST	Créer mouvement	`item_id`, `movement_type`, `quantity`, `reason`, `ticket_id`	`{id, item, quantity}`	✅ Token
`/api/v1/stock/movements/<id>/`	GET	Détail mouvement	-	`{id, item, type, quantity, ...}`	✅ Token
`/api/v1/stock/movements/<id>/`	DELETE	Supprimer mouvement	-	`204 No Content`	✅ Admin
`/api/v1/stock/movements/export/csv/`	GET	Export CSV	Filtres	`text/csv`	✅ Token

8.3 Types d'Articles

Type	Code	Description	Exemples
Pièce détachée	`spare_part`	Pièces de rechange	RAM, SSD, Batteries, Écrans
Consommable	`consumable`	Consommables atelier	Vis, Pâte thermique, Adhésifs

Type	Code	Description	Exemples
Accessoire	`accessory`	Accessoires	Câbles, Adaptateurs, Housses
Outil	`tool`	Outillage	Tournevis, Pinces, Multimètres

---

9. API REST Authentication

9.1 Token & Utilisateurs

Endpoint	Méthode	Description	Params	Response	Auth
`/api/v1/auth/token/`	POST	Obtenir token	`username`, `password`	`{token, user}`	❌ Public
`/api/v1/auth/token/`	DELETE	Révoquer token	-	`204 No Content`	✅ Token
`/api/v1/auth/user/`	GET	Utilisateur courant	-	`{id, username, email, role}`	✅ Token
`/api/v1/auth/user/`	PUT	Mettre à jour profil	`email`, `password`	`{id, username, ...}`	✅ Token
`/api/v1/auth/users/`	GET	Liste utilisateurs	`role`, `is_active`	`[{id, username, email, role}]`	✅ Admin
`/api/v1/auth/users/`	POST	Créer utilisateur	`username`, `email`, `password`, `role`	`{id, username, ...}`	✅ Admin
`/api/v1/auth/users/<id>/`	GET/PUT/DELETE	CRUD utilisateur	-	`{id, username, ...}`	✅ Admin

9.2 Rôles & Permissions

Rôle	Code	Assets	Tickets	Stock	Scanner	Django-Admin
Technicien	`technicien`	R/W	R/W	R	✅	❌
Gestionnaire	`gestionnaire`	R/W	R/W	R/W	✅	❌
Admin CMDB	`admin_cmdb`	R/W	R/W	R/W	✅	❌
Superuser	`superuser`	R/W	R/W	R/W	✅	✅

---

10. Django Admin Système

10.1 URLs Système

URL	Description	Accès
`/django-admin/`	Admin Django système (users, groups, tokens)	Superuser
`/django-admin/auth/user/`	Gestion utilisateurs Django	Superuser
`/django-admin/auth/group/`	Gestion groupes/permissions	Superuser
`/django-admin/authtoken/token/`	Tokens DRF	Superuser
`/django-admin/admin/logentry/`	Logs admin Django	Superuser

10.2 Séparation Admin

```
# config/urls.py
urlpatterns = [
    path('django-admin/', admin.site.urls),      # Système (superusers)
    path('admin/', cmdb_admin.urls),             # Métier IT (techniciens, gestionnaires)
]
```

Admin	URL	Scope	Utilisateurs
Système	`/django-admin/`	Auth, Users, Groups, Tokens, Logs	Superusers uniquement
Métier	`/admin/`	Inventory, Maintenance, Stock, Scanner	Techniciens, Gestionnaires, Admins

---

11. Fichiers Statiques

11.1 CSS

Fichier	Chemin	Description
`admin_base.css`	`/static/admin_cmdb/css/`	Layout, Sidebar, Navbar
`assets.css`	`/static/admin_cmdb/css/`	Module Assets
`tickets.css`	`/static/admin_cmdb/css/`	Module Tickets (Kanban)
`scanner.css`	`/static/admin_cmdb/css/`	Module Scanner
`stock.css`	`/static/admin_cmdb/css/`	Module Stock
`search.css`	`/static/admin_cmdb/css/`	Recherche globale

Fichier	Chemin	Description
`scan_print_hub.css`	`/static/admin_cmdb/css/`	Landing Scan & Print

11.2 JavaScript

Fichier	Chemin	Description
`api.js`	`/static/admin_cmdb/js/`	Axios instance + interceptors
`main.js`	`/static/admin_cmdb/js/`	Layout Vue app (sidebar/navbar)
`assets.js`	`/static/admin_cmdb/js/`	Vue app Assets
`tickets.js`	`/static/admin_cmdb/js/`	Vue app Tickets
`scanner.js`	`/static/admin_cmdb/js/`	Vue app Scanner (ZXing + USB)
`stock.js`	`/static/admin_cmdb/js/`	Vue app Stock
`search.js`	`/static/admin_cmdb/js/`	Recherche globale + autocomplete
`scan_print_hub.js`	`/static/admin_cmdb/js/`	Landing Scan & Print

11.3 Templates

Module	Templates
Base	`admin_base.html`, `admin_login.html`, `dashboard.html`
Assets	`admin/assets/list.html`, `detail.html`, `form.html`
Tickets	`admin/tickets/list.html`, `detail.html`, `form.html`
Scanner	`admin/scanner/index.html`, `print.html`, `templates.html`, `printers.html`, `jobs.html`, `logs.html`
Stock	`admin/stock/list.html`, `detail.html`, `movement.html`
Search	`admin/search/results.html`
Hub	`admin/scan_print/landing.html`
Public	`public/scan_result.html`

11.4 Media

Type	Chemin	Description
QR Codes	`/media/qr_codes/YYYY/MM/`	Images QR générées
Assets Photos	`/media/assets/YYYY/MM/`	Photos assets

Type	Chemin	Description
Stock Photos	`/media/stock/YYYY/MM/`	Photos articles
Print Jobs PDF	`/media/print_jobs/YYYY/MM/`	PDFs impressions batch

---

12. Workflows Complets

12.1 Workflow Réception & Étiquetage

```
┌──────────────────────────────────────────────────────────────────────┐
│              WORKFLOW RÉCEPTION & ÉTIQUETAGE                         │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  1. POST /api/v1/inventory/assets/                                  │
│     └── Asset créé (ex: HP Elitebook G4)                            │
│         │                                                            │
│         ▼                                                            │
│  2. Signal post_save → QRCode.objects.create()                      │
│     └── UUID généré automatiquement                                 │
│         │                                                            │
│         ▼                                                            │
│  3. _generate_qr_image() → media/qr_codes/                          │
│     └── PNG 300x300px généré                                        │
│         │                                                            │
│         ▼                                                            │
│  4. /admin/assets/ → Sélectionner assets                            │
│     └── Action: "Imprimer Étiquettes"                               │
│         │                                                            │
│         ▼                                                            │
│  5. POST /api/v1/scanner/print-labels/                              │
│     └── PDF batch généré (ReportLab)                                │
│         │                                                            │
│         ▼                                                            │
│  6. PrintJob créé → Celery task                                     │
│     └── PDF sauvegardé dans media/print_jobs/                       │
│         │                                                            │
│         ▼                                                            │
│  7. GET /api/v1/scanner/print-jobs/<id>/download/                   │
│     └── PDF téléchargé → Impression MUNBYN                          │
│         │                                                            │
│         ▼                                                            │
│  8. PrintLog enregistré (audit)                                     │
│     └── Qui, quand, quelle imprimante                               │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

12.2 Workflow Scan & Maintenance

```
┌──────────────────────────────────────────────────────────────────────┐
│              WORKFLOW SCAN & MAINTENANCE                             │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  1. /admin/scanner/ → Scanner USB/Webcam                            │
│     └── QR Code scanné → UUID extrait                               │
│         │                                                            │
│         ▼                                                            │
│  2. GET /api/v1/scanner/scan/<uuid>/                                │
│     └── Asset résolu (QR, S/N, ou code interne)                     │
│         │                                                            │
│         ▼                                                            │
│  3. ScanLog enregistré                                              │
│     └── Qui, quand, IP, user_agent                                  │
│         │                                                            │
│         ▼                                                            │
│  4. Fiche asset affichée                                            │
│     └── Nom, S/N, statut, localisation, assigné                     │
│         │                                                            │
│         ▼                                                            │
│  5. Bouton "Créer Ticket"                                           │
│     └── POST /api/v1/maintenance/tickets/                           │
│         │                                                            │
│         ▼                                                            │
│  6. Technicien intervient                                           │
│     └── PATCH /api/v1/maintenance/tickets/<id>/transition/          │
│         │                                                            │
│         ▼                                                            │
│  7. Pièces consommées                                               │
│     └── POST /api/v1/maintenance/tickets/<id>/parts/                │
│         │                                                            │
│         ▼                                                            │
│  8. Stock décrémenté (signal)                                       │
│     └── POST /api/v1/stock/items/<id>/remove-stock/                 │
│         │                                                            │
│         ▼                                                            │
│  9. Ticket résolu → Fermé                                           │
│     └── Historique complet                                          │
│                                                                      │
└──────────────────────────────────────────────────────────────────────┘
```

12.3 Workflow Entrée/Sortie Stock

```
┌──────────────────────────────────────────────────────────────────────┐
│              WORKFLOW ENTRÉE/SORTIE STOCK                            │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ENTRÉE STOCK                                                        │
│  ────────────                                                        │
```

```
│  1. POST /api/v1/stock/items/<id>/add-stock/                    │
│     └── quantity: +50, reason: "Commande fournisseur"          │
│         │                                                       │
│         ▼                                                       │
│  2. StockMovement créé (type: 'in')                            │
│     └── quantity: +50, quantity_before: 20, quantity_after: 70 │
│         │                                                       │
│         ▼                                                       │
│  3. StockItem.quantity mis à jour (signal)                     │
│     └── 20 → 70                                                │
│                                                                 │
│  SORTIE STOCK                                                   │
│  ────────────                                                   │
│  1. POST /api/v1/stock/items/<id>/remove-stock/                │
│     └── quantity: -5, reason: "Réparation", ticket_id: 45      │
│         │                                                       │
│         ▼                                                       │
│  2. StockMovement créé (type: 'out')                           │
│     └── quantity: -5, ticket: #45                              │
│         │                                                       │
│         ▼                                                       │
│  3. StockItem.quantity mis à jour (signal)                     │
│     └── 70 → 65                                                │
│         │                                                       │
│         ▼                                                       │
│  4. PrintLog enregistré (si étiquette imprimée)                │
│     └── Audit complet                                          │
│                                                                 │
└─────────────────────────────────────────────────────────────────┘
```

✅ Checklist Finale

Module	URLs	API Endpoints	Templates	Static Files	Statut
Dashboard	1	0	1	2	✅
Admin Layout	1	0	1	2	✅
Assets	7	12	3	2	✅
Tickets	4	14	3	2	✅
Scanner	7	15	6	2	✅
Stock	4	13	3	2	✅
Search	1	3	1	2	✅
Auth	5	7	1	0	✅
Public	2	1	1	0	✅
TOTAL	**32**	**65**	**20**	**14**	✅

Documentation complète des URLs CMDB Inventory v2.0 — Mars 2026 🎉